# Nguyễn Trần Nhựt Nam

AI Engineer Intern

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## SUMMARY

A fourth-year student passionate about building AI for real-world applications, exploring Machine Learning and Deep Learning, and aspiring to become an AI Engineer who can apply knowledge to practical projects.

## EDUCATION

**Open University TPHCM** *2022 - 2026*

INFORMATiON TECHNOLOGY (10-2022 - 6-2026)

4th-year student

## WORK EXPERIENCE

**Công ty TNHH Baha homes** 3-2026 - 5-2026

**AI Automation Intern**

- Designed and implemented automation workflows to streamline internal operations
- Built AI-driven pipelines for content re-uploading and video generation
- Developed an LLM-based translation system with batch processing to optimize performance
- Designed a RAG system with Smart Connection to improve retrieval accuracy and relevance

## SKILLS

| | |
|---|---|
| **Computer Vision** | CNN, ResNet, Vision Transformer, GAN, Diffusion Models |
| **Deep Learning** | Model training, transfer learning, loss functions, optimization, evaluation metrics, PyTorch training pipeline |
| **RAG** | Faiss, RAG pipeline, rerank, GraphRAG(basic) |
| **Web** | Java Spring, ReactJS, Flask, FastAPI |
| **MLOps** | Docker, MLflow, DVC |
| **Tools** | PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, LangGraph |
| **Soft Skills** | Technical research, teamwork, problem solving, communication |

## PERSONAL PROJECTS

**Agentic AI Workflow & SOP Automation System** 11-2025 - 3-2026

**AI Engineer**

**Technologies / Tools:** Python, LangChain, Pydantic, AsyncIO, Ollama/Groq, Custom Tooling

**Responsibilities:**

- Designed core agents: PlannerAgent, SOPAgent, ExecutorAgent, CRUD Agent.
- Developed a tool-registry system (auto-discovery, grouping, metadata).
- Implemented SOP validation, condition checks, retry logic, and context passing.
- Built static/dynamic execution modes allowing agents to select tools during runtime.
- Integrated LLMs (Ollama/Groq) for reasoning and SOP generation with auto-repair.

**Key Achievements:**

- Delivered a fully functional agentic workflow engine capable of automating complex tasks.
- Reduced LLM JSON errors significantly via strict schema validation + auto-repair loop.
- Enabled flexible tool orchestration and robust multi-step execution with condition logic.

**Github:** nhut-nam/My-AI-Assistant

**Agentic Paper RAG** 5-2026 - Ongoing

**AI Engineer**

**Technologies**: LangGraph, FastAPI, PostgreSQL (pgvector), Ollama, KeyBERT, Sentence-Transformers, Streamlit
**Key Responsibilities**:

- Plan-and-Execute Multi-Agent Architecture: Engineered a LangGraph-based workflow that decouples Planning, Task Execution (via a recursive ReAct research agent), and Answer Synthesis to resolve complex, multi-step queries.
- Multimodal & Hybrid Retrieval: Designed a hybrid search pipeline (Vector + KeyBERT keywords) that retrieves text and processes structural tables/formulas using vision-based OCR to preserve critical academic context.
- Stateful Session Handling: Implemented database-backed conversational state to automatically resolve coreferences or trigger clarification requests for ambiguous queries before routing.

**Github:** nhut-nam/agentic-paper-rag

**Facial Emotion Recognition: EmoNeXt & ConvNeXt with Differential Deep Mutual Learning (FER2013)** 7-2025 - 9-2025

**AI Researcher**

- Built a facial emotion recognition system by combining CNN with Differential Deep Mutual Learning (DiffDML).
- Applied ConvNeXt and EmoNeXt to improve feature extraction capability.
- Achieved 73.5% accuracy — outperforming the original EmoNeXt paper by 0.2%.

**GitHub:** nhut-nam/FacialEmoteRecognition: Build a model for Facial Emotion Recognition

## LANGUAGE

English- intermediate